# ROC-AUC on imbalanced data

Interview notes: concepts, my doubts and the answers, so I can revise without re-asking.

**THE 30-SECOND INTERVIEW ANSWER**

ROC-AUC tells me how well the model ranks positives above negatives. At 0.95, if I pick one random positive and one random negative, the model gives the positive a higher probability 95% of the time. It does not fix a threshold for me. On imbalanced data, though, FPR is a percentage of a huge number of negatives, so a small FPR can still mean thousands of false alarms and very low precision. That is why I also check PR-AUC and precision at my chosen threshold.

## Part 1. What ROC-AUC actually means

**YOUR QUESTION**

*What does a ROC-AUC of 0.80 depict? Explain in simple terms so I don't sound memorized.*

**ANSWER**

Think of it as a fairness test for the model's ranking. The model gives every case a score. A good model should give higher scores to real positives than to real negatives. Run this test many times:

- Pick one real positive case at random.
- Pick one real negative case at random.
- Check which one the model scored higher.

If the positive one wins 80% of the time, ROC-AUC is 0.80. Example: with 100 churners and 900 non-churners, a 0.80 model gives the churner the higher churn probability about 8 times out of 10. A random model would get it right 5 times out of 10 (AUC 0.5).

**HOW TO SAY IT NATURALLY**

ROC-AUC tells me how well the model can tell the two classes apart. If I take one random positive and one random negative, an AUC of 0.8 means the model ranks the positive higher about 80% of the time. So it is about ranking quality across all thresholds, not one specific cutoff.

**YOUR QUESTION**

*What is this 'ranking', and how do I use it in prediction?*

**ANSWER**

A classifier first outputs a probability for each case. Sort all cases by that probability, highest to lowest, and you get a ranked list. A good model puts real positives near the top.

| Customer | Model probability | Actually churned? |
|---|---|---|
| A | 0.92 | Yes |
| B | 0.85 | Yes |
| C | 0.60 | No |
| D | 0.40 | Yes |
| E | 0.20 | No |
| F | 0.05 | No |

D (a real churner) is ranked below C (a non-churner). That is one wrongly ordered pair; ROC-AUC measures how often such pairs come out right.

To make actual predictions you still pick a threshold. At 0.5, only A and B are flagged. At 0.3, A, B, C and D are flagged: recall goes up, precision goes down. Often you do not need yes/no for everyone; you act on the top of the list (for example, call the top 100 likely churners).

> **YOUR QUESTION**
> *So ROC-AUC is not used for threshold selection?*

**ANSWER**

The AUC number is not, but the ROC curve is. The AUC is one summary number used to compare models and judge ranking quality. Each point on the ROC curve is one threshold, so you can pick from it:

- Youden's J: choose the threshold that maximizes TPR - FPR.
- Closest to the top-left corner (FPR = 0, TPR = 1).
- Business constraint: for example 'I need at least 90% recall', then read off the threshold and accept its FPR.

Caveats: on imbalanced data the precision-recall curve is usually better for choosing a threshold, and the final choice should reflect business cost. Choose it on validation data, not the test set.

## Part 2. How ROC-AUC relates to recall, precision and FPR

At every threshold you get one (FPR, recall) pair. Plot all of them and you get the ROC curve; AUC is the area under it.

| Metric | Used in ROC-AUC? | Role |
|---|---|---|
| Recall (TPR) | Yes | y-axis: of real positives, how many did we catch? |
| FPR | Yes | x-axis: of real negatives, how many did we wrongly flag? |
| Precision | No | Not involved at all |

A good model has high recall and low FPR together, so its curve hugs the top-left corner (AUC close to 1). A random model has recall equal to FPR everywhere: a diagonal line, AUC 0.5. Because precision is never used, ROC-AUC cannot see how many false alarms are mixed in with the true catches.

> **YOUR QUESTION**
> *My summary: higher recall means fewer false negatives, so FPR reduces too. Is that right?*

**ANSWER**

Mostly, but one fix: higher recall does not make FPR go down. As you lower the threshold, recall goes up and FPR usually goes up too, because you flag more cases and some are real negatives. A good model achieves high recall and low FPR at the same time (that is what a high AUC means); for a given model, moving the threshold trades one for the other.

## Part 3. Why ROC-AUC struggles on imbalanced data

The AUC is not wrong. The problem is what it hides. Three pieces explain it.

### Piece 1: FPR is a percentage, not a count

FPR = false positives / all real negatives. The same FPR gives very different numbers of false alarms:

- 100 negatives, FPR 5%: 5 false alarms
- 100,000 negatives, FPR 5%: 5,000 false alarms

ROC only shows the percentage, so it cannot tell you whether the false alarms are 5 or 5,000.

### Piece 2: false positives crush precision

Precision = TP / (TP + FP). Fraud example: 100,000 transactions, 100 fraud, 99,900 legit. Recall 90% gives TP = 90. FPR 5% gives FP = about 5,000. Precision = 90 / (90 + 5,000) = about 1.8%. Out of every 100 alerts, only about 2 are real fraud.

| Same ROC point (recall 90%, FPR 5%) | Balanced: 100 fraud, 100 legit | Imbalanced: 100 fraud, 99,900 legit |
|---|---|---|
| True positives | 90 | 90 |
| False positives | 5 | about 4,995 |
| Precision | about 95% | about 1.8% |
| ROC-AUC view | looks great | looks the same, great |

Your ranking statement ('positive ranked above negative 95% of the time') is still correct. But 5% of 99,900 legit cases is about 5,000 transactions that still score higher than a typical fraud case, so fraud cases end up buried in a crowd of legit ones.

**YOUR QUESTION**

*In fraud, recall is the priority and false alarms don't matter. Why does precision come into the picture?*

**ANSWER**

Good question, not a silly one. Two reasons:

- You can cheat recall: flag every transaction as fraud and recall is 100%. Recall alone cannot separate a good model from a useless one.
- False alarms are not free: each alert needs investigation time, legit customers get cards blocked, and if 98 of every 100 alerts are false, the team starts ignoring alerts and misses real fraud anyway.

**THE RIGHT FRAMING**

Recall is the priority, and precision is the operating constraint that keeps the model usable. I would maximize recall subject to a minimum acceptable precision.

### Piece 3: the PR curve shows what ROC hides

The precision-recall curve plots precision against recall across all thresholds; PR-AUC (Average Precision) is the area under it. Because precision uses the raw false-positive count, the curve drops when false alarms pile up. You can read it directly: 'at 90% recall, my precision is only 2%.'

| | ROC-AUC | PR-AUC |
|---|---|---|
| Treats false positives as | A percentage of negatives (FPR) | A count against true positives (precision) |

| | ROC-AUC | PR-AUC |
|---|---|---|
| Sensitive to imbalance? | No, looks good | Yes, reveals the problem |
| Random model scores | 0.5 always | About the positive rate (e.g. 0.001) |
| Question it answers | Can the model separate the classes? | When it raises an alert, is the alert useful? |

## Common mix-ups to avoid

- Precision + recall does not equal 1. They have different denominators (TP+FP vs TP+FN). Pairs that do sum to 1: recall + false negative rate, precision + false discovery rate, specificity + FPR.
- Higher recall does not lower FPR; they usually rise together as the threshold drops.
- ROC-AUC does not use precision.
- The AUC number does not choose a threshold; the curve (ROC or PR) helps you choose one.

## Interview cheat sheet

### Q. What does ROC-AUC of 0.95 mean?

If I pick one random positive and one random negative, the model gives the positive a higher probability 95% of the time. It measures ranking quality across all thresholds.

### Q. Is ROC-AUC used to pick the threshold?

Not the AUC number; it is for comparing models. For thresholds I look at the ROC or, for imbalanced data, the precision-recall curve, and choose based on the business cost of false positives versus false negatives.

### Q. Why does ROC-AUC mislead on imbalanced data?

FPR is normalized by the number of negatives. With millions of legit transactions, a 1-2% FPR still means thousands of false alerts, so precision can be tiny while AUC looks great.

### Q. What would you use instead?

PR-AUC / Average Precision, plus precision and recall at the operating threshold. A random model scores about the positive rate on PR-AUC, so improvement over baseline is visible.

### Q. In fraud, recall matters most. Why care about precision?

Because recall alone can be gamed by flagging everything, and false alerts cost investigation time and customer trust. I maximize recall subject to a minimum acceptable precision.